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Can Technology Solve the Principal-Agent Problem? Evidence from China's War on Air Pollution[†]

By Michael Greenstone, Guojun He, Ruixue Jia, and Tong Liu*

We examine the introduction of automatic air pollution monitoring to counter suspected tampering at the local level, a central feature of China's "war on pollution." Exploiting 654 regression discontinuity designs based on city-level variation in the day that monitoring was automated, we find an immediate and lasting increase of 35 percent in reported PM_{10} concentrations post-automation. Moreover, automation's introduction increased online searches for face masks and air filters, which are strong predictors of purchases. Overall, our findings suggest that the biased and imperfect information prior to automation led to suboptimal investments in defensive measures, plausibly imposing meaningful welfare costs. (*JEL* D82, O13, P28, Q53, Q55, Q58)

Social scientists have long recognized the principal-agent problem inherent in the delegation of authority by governments to bureaucratic officials (e.g., Mitnick 1980; Wilson 1991; Williamson 1996; Aghion and Tirole 1997; and a large literature in public choice). In economics, there exists a rich theoretical literature outlining complicated contracts that align the principal's and agent's incentives (e.g., Laffont and Tirole 1993; Bénabou and Tirole 2006). An alternative and potentially more powerful solution is to find a technology that can greatly reduce the agent's scope for hidden actions.

The execution of environmental policy in China provides an appealing setting to explore these issues. A salient feature in China's political system is that local officials are given high-powered incentives to achieve certain economic and social targets, which link their performance in these targets to their promotion as in a career concerns model (Holmström 1999). While such an incentive system can be effective in achieving targets, it also creates incentives to cheat. The case of air pollution data is an especially poignant example of this dilemma: in recent years, reducing air

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pollution became an important target of the central government (i.e., the principal), yet the power of collecting pollution information is designated to local officials (i.e., the agents). Due to the historically high cost to verify local information and the near-term benefits to high pollution in terms of economic growth, local officials have strong incentives to manipulate air pollution data before reporting them to the central government. The data quality problem is likely to impose significant costs, because it allows for inefficiently high ambient concentration of pollution, causes individuals to undertake inefficient levels of defensives investments, complicates government efforts to undertake international agreements to reduce emissions,[1] and, on the research side, raises questions about the credibility of linking pollution measures to key outcomes such as life expectancy and human capital (Ebenstein et al. 2017; Ebenstein and Greenstone 2020).

This paper examines the introduction of automatic pollution monitoring as a key part of China's extraordinarily successful "war on pollution" (Greenstone et al. 2021). The aims of monitoring automation were to provide reliable measurements of pollution to gauge local officials' success at achieving their targets, to identify where more stringent policy is necessary, as well as to close any gaps between reported concentrations and true concentrations.[2] In this context, automatic monitoring enables real-time sharing of data with the central government and the public and enables improvements in quality assurance checks through statistical cross-validation tools. This change greatly increases the costs for local governments to influence or manipulate the data and provides a compelling context to investigate the efficacy of technology to limit the hidden actions of local officials.

Our analysis exploits several appealing features of the setting. We collect the exact date that automatic monitoring was implemented in 123 cities with 654 monitoring stations, which provides station-specific regression discontinuity (RD) designs to test for manipulation. Moreover, the implementation date varies across cities, allowing for event-study designs that provide a longer-run test for manipulation.

There are two key findings. First, there is striking evidence of the underreporting of air pollution concentrations before automation and improvement in data quality after automation. The RD estimates based on city-level variation on the exact day that monitoring was automated indicate that reported PM_{10} concentrations increased by 35 $\mu g/m^3$ or 35 percent (relative to the post-automation mean of 99.5 $\mu g/m^3$) on average, immediately after monitoring was automated.

By several measures, the estimated increase in reported concentrations is large and economically meaningful. It is almost equivalent to China's remarkable improvement in air quality in the first five years of its "war on pollution" (Greenstone et al. 2021). Based on Ebenstein et al. (2017), a permanent 35 $\mu g/m^3$ difference in PM_{10} concentration implies a loss in life expectancy of approximately 2.2 years for an average person. Additionally, the estimated increase in the *reported* PM_{10}

[1] For example, it was reported that China was reluctant to allow other countries to verify its carbon emission data until the Paris Agreement, where China signed on to an agreement that outlines a single transparent verification system for all countries. See https://www.pri.org/stories/2017-11-08/china-really-stepping-world-s-new-climate-leader.

[2] From our discussion with regulators, we learn that typical ways of manipulation include selected reporting and intentional misreporting. Public media often covers more colorful ways such as spraying water in front of a monitor, which may be less common in routine reporting.

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concentrations meaningfully exceeds the 2019 annual average PM_{10} *levels* for Los Angeles ($20\ \mu g/m^3$) and Boston ($10\ \mu g/m^3$).

We also find that the increase in *reported* PM_{10} concentrations post-automation is evident in difference-in-differences (DiD) designs that exploit the variation in the timing of automation across cities, indicating that the higher recorded concentrations were a longer-run phenomenon. Further, city-specific RD estimates produce quantitative measures of the degree of government misconduct for 76 cities. It is rare to have such measures, and the variation across cities is striking: underreporting is indicated in 33 cities and 12 of them could have underreported the PM_{10} levels by more than $75\ \mu g/m^3$.

The second key finding is that the introduction of automated monitoring led to an immediate and lasting increase in online searches for face masks and air filters. Further, the search responses are stronger in cities that are suspected to be "data manipulating." With respect to the mechanism, it is not possible to isolate whether the changes in internet searches were due to individuals updating their estimates of air pollution concentrations or learning about air pollution more generally due to greater efforts to inform the public about pollution or some combination of both. The absence of city-level data on purchases from the automation period means that we cannot directly measure automated monitoring's impact on defensive expenditures, but we show using data from a different period that searches of face masks and air filters are strongly correlated with their purchases. Overall, our findings suggest that the biased and imperfect information prior to automation led to suboptimal investments in defensive measures, plausibly imposing meaningful welfare costs.

This paper contributes to several strands of literature. First, the Chinese government has been adopting new monitoring and surveillance technologies in many sectors, but little is known about the consequences. While an extensive literature exists on the impacts of technology adoption on economic development, only a few of them have investigated how information and monitoring technology affect public sector governance and efficiency (e.g., Orphanides 2001; Duflo, Hanna, and Ryan 2012; Muralidharan, Niehaus, and Sukhtankar 2016). We provide one example in environmental regulation and the implications are likely to matter in other areas of monitoring and regulation as well.

Second, we contribute to a growing literature on environmental monitoring and regulation (e.g., Duflo et al. 2013, 2018; Shimshack 2014; Greenstone and Hanna 2014; Browne et al. 2019; He, Wang, and Zhang 2020). We find that technology can play an important role in environmental regulation, which matters for researchers, policymakers, and citizens. In a concurrent study, Barwick et al. (2020) also examine the introduction of China's new monitoring system by studying important behavior differences pre-post 2013. The key difference between Barwick et al. (2020) and ours is that we focus on change in information *quality* brought by automated monitoring, while Barwick et al. (2020) treat the introduction of the new system as if pollution information was almost nonexistent before automation and its accessibility was largely the same across China. However, pollution data were available in China before 2013, and we complement their paper by demonstrating a wide variation in pollution data quality across cities. We further show that the increase in internet searches was greater in cities with manipulated data, suggesting that biased information prevented people from optimally protecting themselves against pollution. At the

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same time, Barwick et al. (2020) explore how the new monitoring system affected a wider set of outcomes than this study, most notably the pollution-mortality relationship in China and the pollution-housing price relationship in Beijing.

Third, our study is also related to the literature that assesses the reliability of pollution data from China (e.g., Andrews 2008; Chen et al. 2012; Ghanem and Zhang 2014; and Stoerk 2016). In this literature, researchers find that the distribution of the pre-automation air quality data is not smooth at Air Pollution Index (API) = 100, implying data are manipulated at the politically important threshold. One challenge for these studies is that it is unclear whether data manipulation is local and only exists around the threshold. We show that this concern is more general and researchers should be cautious about China's pre-automation air pollution data.

I. Automating the Air Quality Monitoring System

A. *Policy Motivation*

As China has experienced rapid economic growth over the last several decades, the demand for better air quality and better data on air quality has increased. As recently as the early 2000s, China only provides readings of the opaque API, rather than concentrations of individual pollutants used to construct the API (i.e., PM_{10}, SO_2, and NO_2). A sea change in air pollution reporting was set off when in 2008 the US Embassy in Beijing, and later Consulates in four large Chinese cities (i.e., Shanghai, Guangzhou, Chengdu, and Shenyang), started tweeting hourly $PM_{2.5}$ concentrations readings. These readings were more detailed and typically higher than official Chinese statistics, which led to public doubts about the official readings and elevated concerns about air quality. Further, the Beijing Olympic Games in 2008 and celebrity posts of air quality information and measurement on Weibo (Chinese Twitter) raised public concerns over air pollution and information in China (He, Fan, and Zhou 2016; Ito and Zhang 2020).

To meet the public need and gain public trust, the Chinese government revised the air quality standards in 2012 and later launched the "war on pollution." The Air Quality Index (AQI) was established to replace the API with a stricter standard on PM_{10}. Three more pollutants, including $PM_{2.5}$, O_3, and CO, were added in the determination of the AQI. An automated nationwide monitoring network was established to collect and report pollution information.

B. *What Does Automation Do?*

The automation of China's air quality monitoring network consists of purchasing new monitoring equipment and establishing a new real-time reporting system. New monitors were purchased to monitor $PM_{2.5}$, CO, and O_3, whose information was not available in the past. The equipment and method measuring PM_{10}, SO_2, NO_2 were unchanged, assuring that differences in PM_{10}, SO_2, NO_2 if any, were not due to changes in equipment or measurement method. Instead, the existing equipment was integrated into the new monitoring system. The primary feature of the new system is real-time reporting, which enables online validation and high-standard requirements on measurement.

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Before automation, local environmental bureaus collected data and submitted them to the central authority without validation. This created possibilities for local governments to manipulate the air quality data by, for example, excluding readings from very polluted hours and days, or simply reporting a lower number than was accurate. After automation, opportunities for selective reporting are greatly mitigated as air quality data are sent to the central government in real time. Further, the improvement in surveillance technology allows remote control of data measurement and transmission, as well as quality assurance and control. For example, with the new system, inconsistencies across different monitoring sites that are geographically close would trigger alerts automatically, allowing the central government to further investigate the causes. Additionally, with the availability of real-time data, a higher standard applies to how to measure pollutants. The minimum requirement for calculating daily, monthly, and yearly PM_{10} are respectively increased from 12 to 20 hours per day, 5 to 27 days per month, and 60 to 324 days per year.

With the new system, the concentrations of different air pollutants from more than 1,600 monitoring stations are updated on an hourly basis and are available simultaneously on the Ministry of Ecology and Environment's website (MEE), provincial and municipal environmental bureaus' websites, as well as a large number of mobile apps and third-party websites.

C. *Implementation across Cities*

The automated monitoring system was introduced into different cities in three waves, as planned by the central government (MEE 2017; see online Appendix A1). The variation in timing roughly follows the economic importance of the cities. In the first wave, 74 key polluting cities (496 stations) were required to finish the upgrade by January 1, 2013. These are major cities in China's Capital Economic Zone, the Yangtze River Delta, the Pearl River Delta, and provincial capitals. In the second wave, another 116 cities (449 stations) were ordered to automate by January 1, 2014. These cities include the national key environmental cities predetermined in 2007 and the national model cities for environmental protection (which are evaluated every five years based on their environmental and economic performances since 1997). The third wave further required the remaining 177 cities, which previously did not have systematic air quality monitoring, to build 552 new stations by November 2014.

We focus on 123 cities with 654 monitoring stations that have pollution data both before and after automation. Among these cities, 60 were automated in the first wave and 63 in the second wave. Within a city, all monitors were automated on the same day.

II. Data and Summary Statistics

Our data include measures of monitored air pollution concentrations, weather, aerosol optical depth (AOD), automation dates, and behavioral responses. This section describes these variables and online Appendix A2 reports the summary statistics. We also present descriptive evidence that the quality of air pollution data improved after automation in online Appendices A3 and A4.

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A. *Station Daily Data on Pollutants and Weather*

Before automation, the station-level air pollution data were reported on local environmental bureaus' websites and were continuously collected by us. After automation, real-time pollution data can be accessed through China's Real-Time Air Quality Reporting and Analysis Platform (see Greenstone et al. 2022). Meteorological data are collected from 403 weather stations, which include daily average temperature, precipitation, relative humidity, and wind speed (NOAA 2017). We geocoded all the datasets and matched each pollution monitor with its closest weather station.

B. *Monthly Data on AOD*

AOD data are obtained from two NASA satellites, TERRA, and AQUA with Moderate Resolution Imaging Spectroradiometer (MODIS) (Levy and Hsu 2015). AOD measures the total vertical distribution of particles and gases within a grid (10×10 km) according to the light extinction coefficient. It indicates how much direct sunlight is prevented from reaching the ground by aerosol particles and can be used to infer ground-level pollution, particularly for fine particles such as $PM_{2.5}$, a subset of PM_{10}.[3] The state-of-the-art remote sensing techniques find better correlations between AOD and ground-level PM with coarser spatial and temporal resolutions by month or year (Hoff and Christopher 2009), so we use monthly AOD data.[4] We match the ground-level monitors to AOD grid cells based on coordinates.

C. *Automation Dates across Cities*

We collected news reports on the automation date for each city. As we discussed above, the deadline for automation dates was assigned by the central government; however, cities were allowed to implement the policy before the deadline. Online Appendix A5 plots the distribution of actual automation dates. Among the 60 (63) Wave 1 (Wave 2) cities, 39 (32) automated monitoring at their respective January 1 deadline in 2013 and 2014; in total, 58 percent of Wave 1 and 2 cities automated their monitoring system at their respective deadline. One concern is that the local governments might strategically choose the automation dates to hide underreporting. In the subsequent analysis, we will conduct a separate analysis that focuses on cities that automate at their deadline to assess the possibility that some cities chose to automate before the deadline to hide manipulation.

D. *Behavioral Responses*

We measure behavioral responses through online searches. We focus on Baidu's search indices for "antihaze face masks" and "air filters" (Baidu 2017). Baidu is the biggest search engine in China and provides search indices for specific keywords

[3] PM_{10} is particulate matter 10 micrometers or less in diameter, while $PM_{2.5}$ is particulate matter with a diameter of 2.5 micrometers or less. For context, a human hair is about 100–200 micrometers in width.

[4] AOD at the daily level is a much noisier proxy for PM concentration, because clouds can significantly affect its measurement (e.g., too many missing values).

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that are analogous to Google Trends. The search indices are available from both PC and mobile terminals. We focus on the indices from PCs as the mobile data were not available before May 2013.

The Baidu search index measures city-level search volume for a specific keyword during a specific period. The key advantages of using search indices as the outcomes are that (i) they are available for all the sampled cities both before and after the automation, and (ii) they are strong predictors for actual purchasing behaviors. In fact, the internet search advertising business model is built on the idea that searches are a predictor of sales.

III. Results and Implications

A. *Short-Run Changes in PM_{10}: Evidence from RD Designs*

We use an RD design based on the exact dates of air quality monitoring automation to detect air quality data manipulation:

$$(1)\quad P_{ict} = \beta_1 I(t \geq Auto_{ict}) + \beta_2 f(t - Auto_{ict}) + \beta_3 I(t \geq Auto_{ict}) \times f(t - Auto_{ict}) + \beta_4 W_{ict} + \alpha_i + month_t + u_{ict},$$

where P_{ict} indicates the pollution levels reported by station i of city c at time t (daily/monthly). The term $I(t \geq Auto_{ict})$ is an indicator variable that equals one if station i at time t is automated; $t - Auto_{ict}$ represents the number of days from the automation and is the running variable. We include a "control function," $f(t - Auto_{ict})$, and allow it to differ pre- and post-automation. The station fixed effects (FEs), α_i, account for time-invariant confounders specific to each station. Month FEs, $month_t$, control for seasonality. Weather conditions, W_{ict}, include temperature, precipitation, relative humidity, and wind speed. Finally, u_{ict} is the error term. Since a city can have multiple stations, we cluster our standard errors at the city level.

The parameter of interest, β_1, estimates whether there is a discontinuity in air pollution levels immediately post-automation, after flexible adjustment for the days before/after automation and the covariates. The discontinuity can be estimated by both parametric and nonparametric methods. We emphasize the results from the nonparametric method and use the parametric method as a robustness check.

In the simplest form, we do not include any FEs or control variables in the regression, as the dates of automation are arguably exogenous. To include covariates in the nonparametric RD, we first "residualize" the dependent variable by running an OLS regression in which the dependent variable is *reported* PM_{10} and the explanatory variables are station FEs, month FEs, and weather controls, and then we conduct and RD analysis on the residualized PM_{10}. This procedure provides a consistent estimate of the same RD parameter of interest (Lee and Lemieux 2010).[5]

[5] Alternatively, we include covariates in the nonparametric RD analysis using the methodology developed by Calonico et al. (2019) and obtain similar results.

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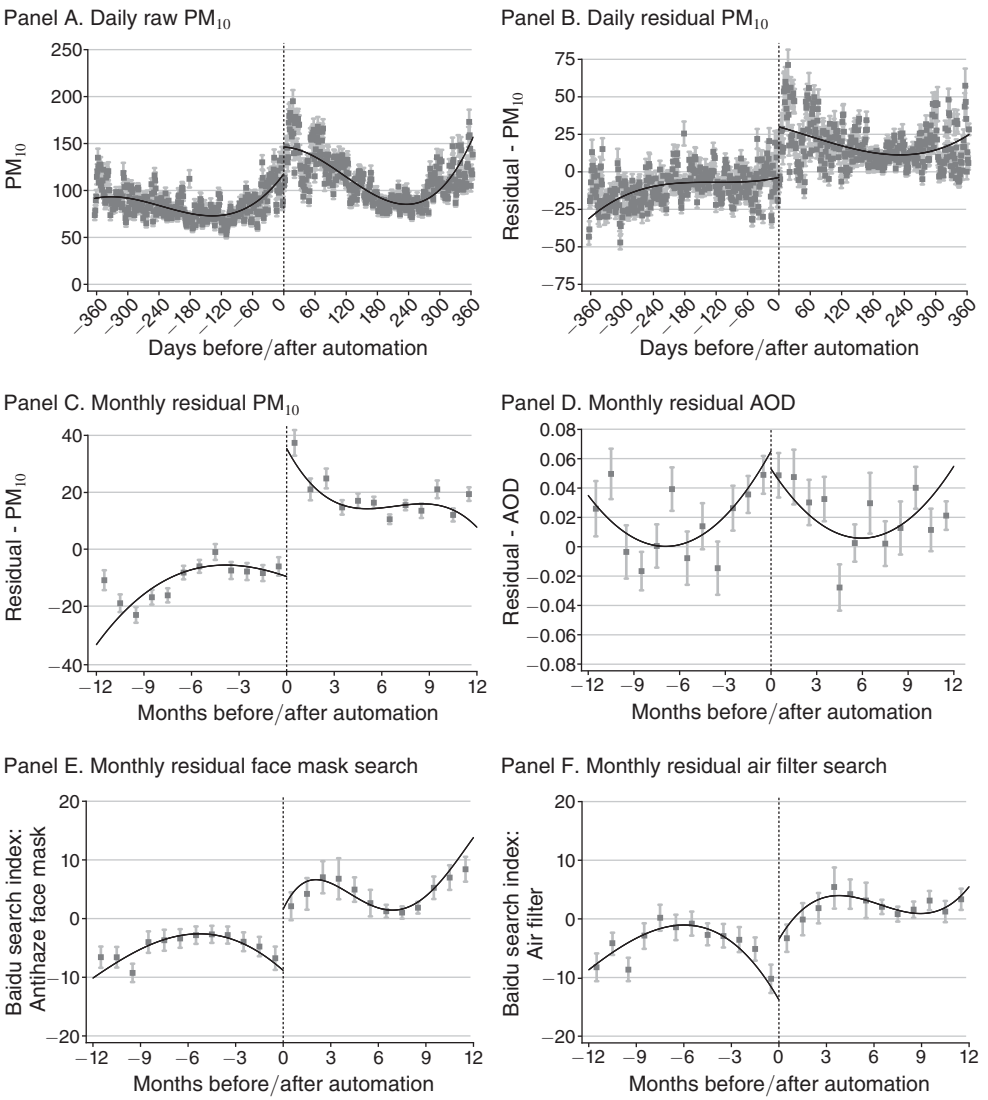


FIGURE 1. RD PLOTS FOR REPORTED PM_{10}, AOD, AND ONLINE SEARCHES

Notes: Panels A–C show the increase in reported PM_{10} immediately after automation using raw daily data, daily residuals, and monthly residuals (absorbing station, month fixed effects, and weather), respectively. Panel D shows no significant change in monthly AOD data. Panels E and F show the increase in monthly online searches for antihaze face masks and air filters after absorbing city fixed effects, month fixed effects, and weather conditions.

We start by visualizing the patterns in the data. In panel A of Figure 1, we plot the raw daily *reported* PM_{10} concentration data. Panel B of Figure 1 plots the residualized concentrations after adjustment for monitoring station FEs, month FEs, and meteorological conditions. In both panels, we observe a striking increase in *reported* PM_{10} immediately after automation. Similar pattern can be observed using monthly PM_{10} data, as shown in panel C of Figure 1. The estimated increase in *reported* PM_{10} concentrations on the day of automation is 35 $\mu g/m^3$ in panel B of Figure 1 with daily data, and 38 $\mu g/m^3$ with monthly data in panel C of Figure 1.

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Table 1—Automating Air Quality Monitoring System and Reported PM_{10}

	(1)	(2)	(3)	(4)	(5)
Panel A. RD estimates					
RD in PM_{10} (daily)	34.7	34.9	27.5	64.7	57.1
	(10.7)	(5.8)	(9.8)	(9.9)	(8.6)
RD in AOD	0.065	−0.005	0.026	−0.030	−0.003
	(0.044)	(0.021)	(0.031)	(0.029)	(0.025)
Sample	All	All	Wave 1	Wave 2	Deadline
Station fixed effects		Y	Y	Y	Y
Month fixed effects		Y	Y	Y	Y
Weather controls		Y	Y	Y	Y
Observations (daily)	91,470	232,326	81,950	68,456	86,042
Bandwidth (days)	109	263	140	234	184
Observations (monthly)	5,057	5,851	3,173	2,316	4,894
Bandwidth (months)	6	7	7	6	10
Panel B. Event-study estimates					
	PM_{10}	PM_{10}	PM_{10}	PM_{10}	$\log(PM_{10})$
7–12 months before	−8.5	−17.2	−10.7	−10.8	−0.13
	(4.7)	(6.7)	(7.7)	(9.7)	(0.07)
5–6 months before	6.8	−19.2	10.5	−2.2	0.02
	(6.0)	(9.3)	(8.5)	(12.1)	(0.11)
3–4 months before	−6.4	−12.0	−2.8	−5.2	−0.03
	(5.6)	(6.9)	(7.3)	(9.2)	(0.09)
1–2 months after	60.3	31.4	66.5	45.6	0.24
	(11.0)	(11.1)	(14.3)	(16.3)	(0.09)
3–4 months after	45.0	33.6	47.2	32.5	0.32
	(7.8)	(8.8)	(10.7)	(14.2)	(0.12)
5–6 months after	28.1	22.2	33.4	29.0	0.29
	(6.7)	(8.0)	(9.7)	(13.7)	(0.12)
7–12 months after	40.0	9.8	42.9	15.8	0.24
	(6.1)	(8.8)	(7.7)	(14.0)	(0.15)
Sample	Deadline	Deadline	+Matching	+Matching	+Matching
Weather controls	Y	Y	Y	Y	Y
Station fixed effects	Y	Y	Y	Y	Y
Month fixed effects	Y		Y		
Year-month fixed effects		Y		Y	Y
R^2	0.34	0.35	0.33	0.34	0.38
Observations	176,426	176,426	186,499	186,499	186,469

Notes: In panel A, each cell represents a separate nonparametric RD estimate. Triangle kernel is used and optimal bandwidth is selected by Calonico, Cattaneo, and Titiunik's (2014) method. Columns 1 and 2 use the entire sample to estimate the discontinuities; there are 1,049,325 daily observations before bandwidth selection. Columns 3 and 4 use the Wave 1 and Wave 2 cities. Column 5 uses cities that automated the monitoring system at their deadlines. In panel B, we use data from January 1, 2012 to December 31, 2013 for the event-study estimates. The omitted period is "1–2 months before automation." In columns 1 and 2, there are 242 Wave 1 (deadline) stations and 123 Wave 2 (deadline) stations. In columns 3 to 5, each Wave 1 (deadline) station is matched with its nearest Wave 2 (deadline) station with replacement. Weather controls include temperature, relative humidity, precipitation, and wind speed. Standard errors clustered at the city level are reported in parentheses below the estimates.

We present RD estimates from equation (1) in panel A of Table 1. Columns 1 and 2 report the results from the local linear RD with and without covariates (i.e., W_{ict}, α_i, $month_t$). The bandwidths are 109 days and 263 days for PM_{10} in the two columns, respectively, which are the optimally selected by Calonico, Cattaneo, and Titiunik's (2014) method. Both columns use a triangle kernel weighting function. The estimated discontinuity is around 35 $\mu g/m^3$, corresponding to a 35 percent increase, relative to the overall post-automation mean (99.5 $\mu g/m^3$).

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The remaining columns analyze subsets of the monitors. In columns 3 and 4, the *reported* PM_{10} levels increased 28 $\mu g/m^3$ (or 33 percent) for Wave 1 cities and 65 $\mu g/m^3$ (or 76 percent) for Wave 2 cities (see online Appendix B1 for the RD plots). In column 5, we restrict the sample to the 71 cities (58 percent of the sample) that implemented the policy at the deadline (drawn from both waves) to explore the possibility that local governments strategically choose their automation dates to hide underreporting. It is at least plausible that cities that automated at their deadline are less likely to have strategically chosen the automation dates to hide manipulation. We find an increase of 57 $\mu g/m^3$ in *reported* PM_{10} concentrations for this subgroup, which is larger than in the full sample but qualitatively similar. We conclude that although there may have been some strategic choice of automation dates, the influence of such behavior on the overall results is at most modest.

We provide several sets of results that together lend additional credibility to the baseline findings. First, we find that all the weather variables are continuously distributed across the threshold (reported in online Appendix B2), suggesting that the dramatic changes in the *reported* PM_{10} levels were not driven by weather conditions. Second, alternative kernel weighting methods and the parametric RD approach yield similar estimates, as reported in online Appendix B3. Finally and most importantly, we fit equation (1) for monthly AOD and find no discontinuity in AOD levels after automation (panel D of Figure 1, panel A of Table 1, and online Appendix B3), confirming that this measure of true air quality did not deteriorate after the automation.[6]

We then explore the heterogeneity in data quality across different cities, which provides a rare opportunity to learn about the degree of manipulation at the city level. Specifically, we estimate equation (1) city by city using the nonparametric approach, with the unit of observation being a monitor by day but restricting the effect of automation to be equal across monitors within a city. We note a change in the sample, relative to the 123-city (654-monitor) primary sample. Some cities suspended data reporting while they installed and tested the new automatic monitors for $PM_{2.5}$, CO, and O_3. As a result, 47 cities did not report PM_{10} readings for more than two months preceding the initiation of automatic monitor reporting and it is therefore challenging to credibly apply the RD approach to these cities individually. We drop these cities for this analysis, leaving a sample of 76 cities (464 monitors).[7]

Figure 2 plots the city-specific RD estimates and their 95 percent confidence intervals. The RD coefficient is positive for more than 70 percent of these cities. Among them, 33 estimates are positive and statistically significant at the 5 percent level. These results suggest that our baseline findings in panel A of Table 1 are driven by widespread manipulation, rather than a few cities. The average of all the RD coefficients is 28.2 $\mu g/m^3$ (the red line). We observe substantial variation in the estimated discontinuities. Particularly noteworthy findings are that 12 cities have discontinuities greater than 75 $\mu g/m^3$ and 11 of them are statistically significant at the 1 percent level. Online Appendix B5 plots the spatial variation in manipulation.

Online Appendices B6–B10 provide a battery of additional evidence that data quality improved after automation. Specifically, we find that the variability of

[6] An alternative measure of actual air quality is the $PM_{2.5}$ and AQI readings from US consulates in five Chinese cities. We do not observe significant RD in US consulate data, too.

[7] Excluding the 47 cities does not affect our baseline RD estimates, as shown in online Appendix B4.

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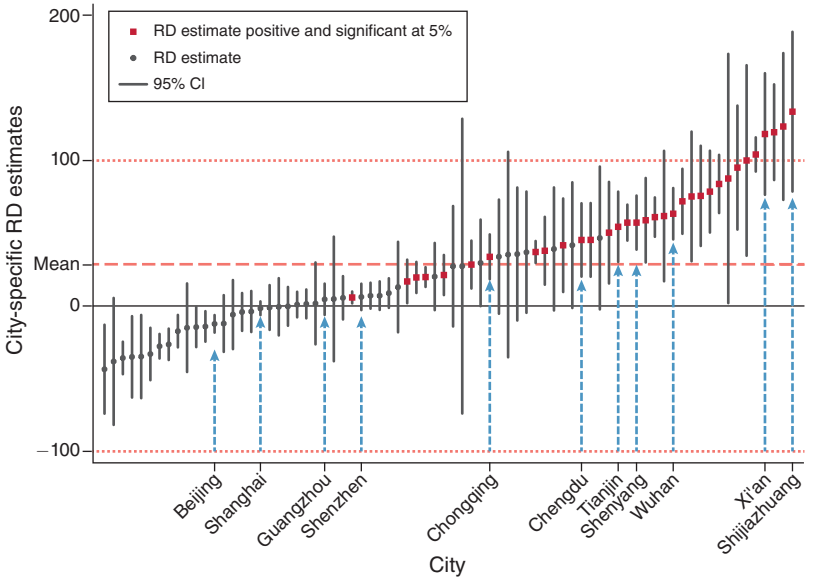


FIGURE 2. MAGNITUDES OF MANIPULATION IN CHINESE CITIES

Notes: The RD estimates and their 95 percent confidence intervals are plotted for 76 cities. The average RD estimate is denoted by the red dashed line.

reported PM_{10} became larger post-automation (online Appendix B6), the discontinuities in *reported* SO_2 and NO_2 (both of which play only small roles in determining promotion for local officials) are small (online Appendix B7), the data quality change is not simply driven by tighter data collection requirement (online Appendix B8), the distribution of post-automation PM_{10} data is smooth at different critical thresholds in the air quality standards (online Appendix B9), and the correlation/partial-correlation between *reported* PM_{10} and AOD became stronger after automation (online Appendix B10).

B. *Medium-Run Changes in Reported PM_{10}: Difference-in-Differences Estimates*

The RD approach offers a demanding test of the effect of automation immediately after its implementation. We complement the main analysis using a set of difference-in-differences models that provide less strict tests but offer the potential to estimate the medium-run effects of automation. Specifically, we conduct event-study analyses using the following equation:

$$P_{ict} = \gamma_\tau \sum_{\tau=-4}^{+3} Auto_{ic\tau} + \beta W_{ict} + \alpha_i + month_t + \epsilon_{ict}, \tag{2}$$

where P_{ict} and W_{ict} are defined in the same way as the RD setup above. Here, α_i are the station FEs and $month_t$ are the month or year-by-month FEs (we will show results for both specifications). Values of $Auto_{ic\tau}$ indicate different periods before and after the automation, and we set the pollution readings 1–2 months before the

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automation date as the reference group ($\tau = -1$). Then, $\tau \in \{-4, -3, -2, 0, 1, 2, 3\}$ respectively refers to 7–12 months, 5–6 months, and 3–4 months before automation, and 1–2 months, 3–4 months, 5–6 months, and 7–12 months after automation.

The coefficients of γ_0 to γ_3 allow us to examine whether the automation increases *reported* PM_{10} readings in the short and medium runs (relative to the PM_{10} readings 1–2 months just before automation). The coefficients of γ_{-4} to γ_{-2} further tell us if the PM_{10} readings months ago were comparable to the baseline readings (PM_{10} readings 1–2 months before automation).

To avoid composition changes in this dynamic analysis, we restrict the sample to cities that automated their monitoring stations only at the deadline of their respective wave and focus on the period from January 1, 2012 to December 31, 2013. Thus, the "treatment" monitors are from Wave 1 cities where automation occurred on January 1, 2013, and the "control" monitors are from Wave 2 cities where automation never occurred during this two-year period. With this setup, the "control" monitors are never treated and provide a plausibly credible counterfactual for the "treatment" monitors. Further, this is one approach to confronting the challenges associated with the staggered assignments of treatment.[8]

The results are reported in panel B of Table 1. There is some evidence of increasing *reported* PM_{10} concentrations prior to automation in columns 1 and 2 specifications, which raise questions about the DiD's identifying assumption. To mitigate the influence of any differences in pre-trends, the columns 3–5 specifications match each monitoring station in the Wave 1 (deadline) cities to its (geographically) nearest monitoring station in the Wave 2 (deadline) cities with replacements and reestimate equation (2) using this paired sample. The idea is that geographically adjacent pairs of monitors should have similar pre-automation trends in PM_{10} readings. Indeed, after matching, the coefficients for all the "lead" variables become small in magnitude and statistically insignificant.

We find that *reported* PM_{10} concentrations are substantially higher post-automation. While the increase in levels generally declines over time (see columns 1–4), it is relatively stable when the natural logarithm of reported PM_{10} is the dependent variable, in which the coefficients are approximations to percentage changes (24 percent to 32 percent in column 5). This difference in results is largely due to seasonality in China's pollution concentrations: the first two months post-automation are in the winter season, which is the time of year when pollution concentrations are the highest in China. Panel A of Figure 3 presents the column 5 results graphically.

Overall, we conclude from the previous subsection and this one that automation led to immediate and sustained increases in *reported* PM_{10} concentrations.

C. *Correcting Pre-Automation PM_{10} Data*

As a by-product of this paper, we attempt to correct the pre-automation PM_{10} data using machine learning. The intuition is that we can train prediction models based on the post-automation AOD-PM_{10} relationship and use them to predict pre-automation

[8] An emerging literature shows that estimates from two-way FE models with staggered treatment assignments are difficult to interpret. See discussions in de Chaisemartin and D'HaultfŒuille (2018, 2019); Goodman-Bacon (2018); Imai and Kim (2021).

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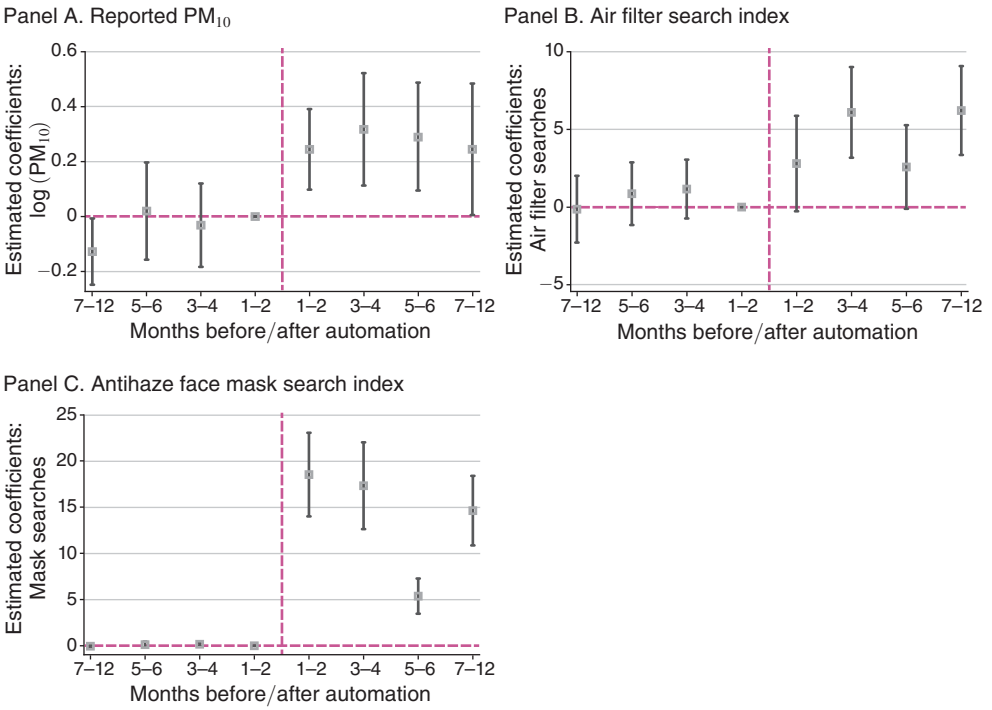


Figure 3. Event-Study Estimates for Reported PM_{10} and Online Searches

Notes: This figure shows the event-study estimates and their 90 percent confidence intervals for reported PM_{10} and online search indices. Panel A corresponds to the regression results reported in column 5 of panel B in Table 1. Panels B and C correspond to the regression results reported in columns 2 and 4 of panel B in Table 2. In panels B and C, the treatment group consists of cities that automated the system on January 1, 2013, and the control group consists of cities that automated the system on January 1, 2014. Panel A matches the treatment stations with the nearest control stations. We keep data from January 1, 2012 to December 31, 2013 for the estimations and omit the "1–2 months before automation" period in the event-study regressions. Location fixed effects, month fixed effects, and weather conditions are controlled.

PM_{10} levels. In online Appendix C, we train an artificial neural network to correct the pre-automation PM_{10} data and provide the data as an online Appendix for other researchers (see Mullainathan and Spiess 2017 for the method). The correction shifts the distribution of the pre-automation PM_{10} data to the right. The mean of PM_{10} in this corrected distribution is 27.3 $\mu g/m^3$ or 32 percent higher than the mean of the reported pre-automation distribution.

D. *Behavioral Responses*

This subsection tests for behavioral responses to the improvements in the quality and availability of pollution information. We focus on city-level search indices for "antihaze face masks" and "air filters" in the period before and after automation. Comparable data on purchases would allow for a revealed preference estimate of the welfare consequences of automation, but such data were unavailable during our study period. However, we present evidence of a strong relationship between searches and purchases in online Appendix D1, albeit from a different period and

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set of cities.[9] Here, we document that a 10 percent increase in the city-level antihaze face mask (air filter) search index corresponds to a 3.1 percent (6.0 percent) increase in city-level sales using Taobao sales data (Sun, Kahn, and Zheng 2017).

Turning to the search data, panels E and F of Figure 1 present the RD plots for the two outcomes, respectively. Panel A of Table 2 summarizes the RD estimates using equation (1). We find that monthly online searches for "face masks" immediately tripled (columns 1–2) after automation and searches for "air filters" increased by 17–20 percent.

We then examine the differences in people's responses between data-manipulating cities and normal cities. For the 76 cities with RD estimates, we define a city as a data-manipulating one if its RD estimate is positive and statistically significant at 5 percent level. For the remaining cities that have missing data issues, we compare *reported* PM_{10} levels between January–June 2013 and January–June 2014: if the *reported* average PM_{10} concentrations increased by 35 $\mu g/m^3$ in the city, we define it a data-manipulating city (13 cities meet this criterion).[10] Column 3 reports the results using data from the "normal" cities and column 4 focuses on "data-manipulating" cities. In "data-manipulating" cities, the post-automation increase in searches was substantially larger: more than four times for "face masks" and 28 percent higher for "air filters." While it is tempting to interpret these larger estimates in column 4 as being entirely due to individuals' learning that actual PM_{10} concentrations were higher than they had believed, it is also possible that the increase in news about air pollution disproportionately increased in these cities at the same time.

Panel B of Table 2 and panels B and C of Figure 3 report the event-study estimates using equation (2). We find that the higher rates of searches for these two terms were sustained and still evident 7–12 months after automation.[11] It is reassuring that the RD and event-study estimates are similar in size. These results imply automation leads to an immediate and sustained increase in the purchases of goods that protect individuals from PM_{10}.

Because most cities adopted the new technology just on two dates (January 1 of 2013 and 2014), a legitimate concern is that there might be other policies launched at the beginning of a new year that could drive the behavioral changes. This concern is mitigated by the fact that the event-study estimates persist. We investigate this issue in two additional ways. First, we conduct similar analyses around January 1, 2015 in online Appendix D2 and find no significant behavioral changes. Second, we report the RD estimates separately for cities that automated their systems at their deadline and cities that automated their systems before their deadline in online Appendix D3. We find that the behavioral responses can be observed in both groups, even though the effect size is greater in the deadline cities.[12]

[9] The Taobao sales data are available in 34 Wave 1 cities (4 municipalities and 27 provincial capitals) from April 2013 to April 2014.

[10] We use 35 $\mu g/m^3$ as the threshold to define the data-manipulating status for these cities because 35 $\mu g/m^3$ is the average discontinuity in Table 1. In reality, it is almost impossible for a city to reduce its PM_{10} concentration by 35 $\mu g/m^3$ within a year. Slightly changing the definition of data-manipulating city, for example, using a more/less stringent definition, does not affect our results in Table 2.

[11] The searches for antihaze masks were very low before automation, since antihaze became an issue only after automation.

[12] A limitation of this comparison is that pollution levels are higher in winter and the deadline is January 1, which is in the midst of winter. Consequently, it is possible that the larger response in internet searches in

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Table 2—Automating Air Quality Monitoring System and Avoidance Behaviors

	(1)	(2)	(3)	(4)
Panel A. RD estimates				
RD in face mask searches	10.10	11.03	6.46	18.77
(preautomation mean = 0.62)	(1.58)	(1.66)	(2.03)	(2.63)
RD in air filter searches	7.36	8.73	5.16	16.30
(preautomation mean = 35.5)	(3.60)	(1.86)	(2.50)	(2.39)
RD in log (face mask searches + 1)	1.06	1.15	0.90	1.72
(preautomation mean = 0.16)	(0.17)	(0.17)	(0.21)	(0.25)
RD in log (air filter searches + 1)	0.18	0.16	0.12	0.25
(preautomation mean = 3.30)	(0.10)	(0.04)	(0.05)	(0.06)
Sample	All	All	Normal	Manipulate
City fixed effects		Y	Y	Y
Month fixed effects		Y	Y	Y
Weather controls		Y	Y	Y
Panel B. DiD estimates				
	Mask searches	Mask searches	Filter searches	Filter searches
7–12 months before	0.00	−0.05	−0.15	−0.14
	(0.00)	(0.09)	(1.31)	(1.31)
5–6 months before	0.00	0.12	0.62	0.86
	(0.00)	(0.20)	(1.16)	(1.23)
3–4 months before	0.00	0.17	0.91	1.16
	(0.00)	(0.11)	(1.12)	(1.15)
1–2 months after	18.60	18.52	2.87	2.80
	(2.78)	(2.75)	(1.90)	(1.87)
3–4 months after	17.39	17.31	6.11	6.10
	(2.86)	(2.86)	(1.78)	(1.78)
5–6 months after	5.43	5.37	2.48	2.58
	(1.19)	(1.16)	(1.64)	(1.64)
7–12 months after	14.45	14.62	6.00	6.22
	(2.31)	(2.29)	(1.72)	(1.74)
Sample	Deadline	Deadline	Deadline	Deadline
City fixed effects	Y	Y	Y	Y
Year-month fixed effects	Y	Y	Y	Y
Weather controls		Y		Y
R^2	0.32	0.32	0.53	0.53
Observations	51,901	51,900	51,170	51,169

Notes: In panel A, each cell represents a separate RD estimate. Triangle kernel is used and optimal bandwidth is selected by Calonico, Cattaneo, and Titiunik's (2014) method. Columns 1 and 2 use the entire sample from the 123 cities, consisting of 8,661 mask search observations and 8,590 filter search observations before bandwidth selection. Column 3 limits the sample to "normal" cities where we fail to detect manipulation and column 4 focuses on data-manipulating cities defined in Section IV. In panel B, we use data from January 1, 2012 to December 31, 2013 for the event-study estimates. The omitted period is "1–2 months before automation." The 39 Wave 1 deadline cities are the treatment group and the 32 Wave 2 deadline cities are the control group. Weather controls include temperature, relative humidity, precipitation, and wind speed. Standard errors clustered at the city level are reported in parentheses below the estimates.

We do not believe that it is possible to neatly isolate whether the post-automation behavioral changes were due to individuals updating their estimates of air pollution concentrations, or learning about air pollution more generally, or some combination of both. Regardless, it seems reasonable to conclude that the biased and

deadline-complying cities reflects greater awareness of pollution and responsiveness to information when concentrations are the highest.

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imperfect information about air pollution imposed meaningful welfare costs prior to automation.

IV. Conclusion

Governments delegate authority to bureaucratic officials, which makes the principal-agent problem inherent to government organizations. The case of pollution data quality in China shows that high-powered incentives in the public sector can be a double-edged sword: when local officials obtain a strong incentive to perform better, they also have incentives to manipulate data.

The advancement of information technology and the adoption of real-time monitoring offer a possible tool to address this downside. We show that automating the monitoring system significantly improves data quality. The improvement of data quality is an important underlying factor to explain China's success in its "war on pollution" in recent years—it is difficult to imagine an effective policy without reliable information. Besides, the more reliable information made available post-automation seems to have induced more people to take avoidance behaviors against pollution, which implies welfare gains that are of a potentially significant magnitude. That being said, new monitoring and surveillance technologies are likely to have other important implications for governance about which there is much to learn. Our study is just one example of the consequences of technological advancement for governance, and we believe that this is a rich area for research going forward.

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